Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with an MS in CS from USC and 2+ years building backend and data systems in **Python**, TypeScript, and SQL, with **security** research in malware URL detection, authentication, and **data-driven** threat analysis at scale.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data pipelines** across PostgreSQL, SQL Server, and cloud storage with incremental loads, validation, and reconciliation checks, running 5K+ monthly jobs at 99%+ production reliability for 500+ users.
- Enforced tenant isolation with Azure SQL **row-level security** and rebuilt async job processing with failure isolation, tripling throughput across 1K+ AI jobs and cutting dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack React and Flask platform with **secure file ingestion** for PDF, DOCX, and ZIP uploads and Flask-Login authentication, serving 150+ instructors and 10K+ daily requests.
- Designed concurrent batch processing for 1K+ files per run with fault-tolerant execution and a provider-agnostic **AI** layer across GPT-4o, Claude, and Qwen-VL, cutting average feedback turnaround by 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built cross-platform data-sharing software on a **microservice architecture** for Windows, Linux, and macOS with automated VM builds, installs, and 20+ unit and integration test suites in nightly Jenkins.
- Consolidated user activity logs from multiple sources into normalized formats with AWS Glue and S3 **ETL**, boosting pipeline efficiency 68%, and optimized APIs with caching to improve response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight TypeScript and Node.js microservices with React, including JWT authentication and **role-based access control**, synchronizing 2M+ records across multiple locations through REST APIs and real-time WebSockets.
- Integrated Snyk and OWASP Dependency-Check **vulnerability scanning** into Jenkins CI/CD with approval gates, and secured Azure SQL data with encryption, firewalls, and access controls, reaching 95%+ build validation across releases.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Built a **malicious URL detection** pipeline over 650K+ URLs, engineering 21 behavioral indicators such as IP presence, URL length, and suspicious keywords to classify phishing and malware threats.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN models, reaching 96%+ **threat detection** accuracy and reporting actionable insights into phishing and malware trends.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
Backend & Frontend: FastAPI, Flask, Node.js, REST APIs, WebSockets, Microservices, React, Next.js
Security: Malware Detection, Phishing Detection, Authentication, JWT, RBAC, Row-Level Security, Snyk, OWASP, Encryption
Data & ML: ETL Pipelines, PostgreSQL, SQL Server, MongoDB, Databricks, Spark, PyTorch, Feature Engineering, Random Forest, LSTM
Infrastructure: Linux, Docker, Kubernetes, AWS, Azure, CI/CD, Jenkins, GitHub Actions
Fundamentals: Data Structures, Algorithms, Operating Systems, Distributed Systems

Projects

On-Prem Security Telemetry Lake | *Python, Kafka, ClickHouse, FastAPI, React, Docker*

- Designed a self-hosted **security telemetry** pipeline that streams firewall, DNS, and endpoint logs through Kafka into ClickHouse, scores anomalies with detection rules and ML, and surfaces alerts in React.

Comprehensive Analysis of Blockchain for User Authentication Paper | *Authentication, RBAC, Threat Modeling*

- Co-authored an IEEE paper on blockchain **authentication** for users and IoT devices, threat-modeling identity theft, replay, man-in-the-middle, and phishing attacks against centralized identity stores and smart-contract verification.

A Detailed Exploration to Quantum Resistant Blockchain Technology Paper | *FALCON, Post-Quantum Cryptography*

- Co-authored an IEEE paper proposing a **post-quantum** blockchain framework with FALCON lattice-based signatures, benchmarked at 957.8 TPS and 224ms latency against an ECDSA baseline of 968.3 TPS at peak load.